THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** independent variable	The quantity changed by the investigator is the independent variable.
2	**(b)** Robert Hooke	Hooke saw box-like compartments and called them cells.
3	**(b)** physical, mental and social well-being	Health is more than the absence of disease.
4	**(b)** cell or battery	A cell converts chemical energy to electrical energy.
5	**(c)** push or a pull	Forces are pushes or pulls.
6	**(c)** force acting per unit area	P = F / A.
7	**(b)** tiny particles	Matter consists of very small particles.
8	**(b)** element	That is the definition of an element.
9	**(c)** solution	Uniform composition.
10	**(c)** lie in the same plane	All three are coplanar.
11	**(c)** rotation of the Earth on its axis	Earth spins from west to east.
12	**(b)** living and non-living components interacting with each other	Biotic and abiotic parts together.
13	**(c)** water, air and a suitable temperature	Conditions needed for life are present.
14	**(d)** a question or curiosity about an observation	Questions drive investigations.
15	**(d)** cell, tissue, organ, organ system, organism	Cells form tissues; tissues form organs, and so on.
16	**(d)** something a doctor can observe or measure, such as a rash	Signs are observable.
17	**(b)** nichrome	Nichrome has high resistance and a high melting point.
18	**(c)** rough surfaces	Rough surfaces interlock.
19	**(d)** barometer	Barometer.
20	**(c)** melting	Fusion.
21	**(c)** malleability	Malleable.
22	**(d)** brass	Alloy.
23	**(a)** diffuse reflection	Irregular.
24	**(d)** 27 days	Orbital period of the Moon.
25	**(d)** food chain	Linear sequence.
26	**(c)** nitrogen	About 78 per cent.
27	**(a)** length of the pendulum	Length is the quantity that is changed.
28	**(a)** vacuole	A large vacuole stores water and substances.
29	**(d)** vitamin C	Bleeding gums and weak joints.
30	**(a)** a resistor	Resistor symbol.
31	**(b)** positive and negative	Two kinds of charge.
32	**(b)** very low pressure	Low pressure centre.
33	**(c)** increase in temperature	Warmer liquids evaporate faster.
34	**(a)** a combination of two or more substances not chemically combined	Components keep properties.
35	**(d)** (mass of solute / mass of solution) x 100	Standard definition.
36	**(a)** R = 2f	Focus is midway.
37	**(c)** solar calendar	Gregorian.
38	**(b)** decomposition	Recycling.
39	**(d)** a small part of the total water	About 3 per cent is fresh and little is accessible.

40	**(d)** dependent variable	It is the quantity that is measured as the result.
41	**(a)** microscope	A microscope magnifies tiny things.
42	**(d)** getting along well with family and neighbours	Social health concerns relationships.
43	**(b)** a gap in the circuit	An open switch breaks the path.
44	**(d)** newton	Named after Isaac Newton.
45	**(d)** pascal	1 Pa = 1 N/m squared.
46	**(d)** sugar dissolves in water without raising the water level much	Sugar particles occupy spaces between water particles.
47	**(a)** O_2	Oxygen contains atoms of only one kind.
48	**(c)** solute	Substance dissolved.
49	**(b)** angle of incidence equals angle of reflection	i = r.
50	**(d)** 24 hours	One day.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** sunlight	Non-living.
2	**(b)** crust	Thin solid layer.
3	**(b)** a flexible approach that may be repeated and revised	Real investigations loop back and revise earlier steps.
4	**(b)** tissue	A tissue is a group of similar cells.
5	**(d)** chronic	Chronic diseases are long lasting.
6	**(b)** low melting point	It melts and breaks the circuit during overload.
7	**(d)** lubricants	Oil forms a layer.
8	**(c)** wind	Wind is air in motion.
9	**(b)** evaporation	Surface process.
10	**(c)** ductility	Ductile.
11	**(b)** nitrogen	Most abundant gas.
12	**(b)** regular reflection	Parallel beam stays parallel.
13	**(a)** 29.5 days	Synodic period.
14	**(b)** grass, grasshopper, frog, snake	Producer first.
15	**(a)** 21 per cent	One-fifth.
16	**(b)** check the method and repeat the experiment	Differences must be investigated.
17	**(a)** give shape and protection to plant cells	It is rigid and strong.
18	**(a)** vitamin A	Needed for vision.
19	**(d)** a broken connection or a fused filament	Open circuit.
20	**(c)** repel each other	Same charges repel.
21	**(c)** eye	Clear and calm.
22	**(d)** spread out in a windy place	More surface and wind.
23	**(d)** in a fixed ratio by mass	Law of constant composition.
24	**(a)** 20 per cent	20/100 x 100.
25	**(d)** convex	Wide field of view.
26	**(c)** lunar calendar	Months follow the Moon.
27	**(b)** carbon dioxide	CO2 + water.
28	**(a)** ice caps and glaciers	Frozen.
29	**(d)** thickness of the rolled dough	All factors other than the one being studied are kept constant.
30	**(b)** nucleus	The nucleus holds the genetic material and directs cell activities.
31	**(c)** communicable	Communicable diseases are infectious.
32	**(d)** from the positive terminal through the circuit to the negative terminal of the cell	Conventional current is from the positive terminal.
33	**(b)** the speed, direction or shape of an object	Effects of force.
34	**(a)** 50 Pa	100 / 2 = 50 Pa.
35	**(d)** constantly moving	Particles are always in motion.
36	**(c)** CO_2	Carbon dioxide has two kinds of atoms chemically joined.
37	**(d)** solvent	Dissolving medium.
38	**(d)** 35 degrees	i = r.
39	**(a)** 365 and a quarter days	One year.
40	**(d)** bacteria	Living.

41	**(d)** mantle	Thickest layer.
42	**(d)** revise the hypothesis and test again	Honest reporting is essential.
43	**(b)** an organ	It is made up of several tissues working together.
44	**(a)** tuberculosis	TB can last for months.
45	**(a)** tungsten	Tungsten has a very high melting point.
46	**(a)** less	Rolling friction is smaller than sliding friction.
47	**(b)** differences in air pressure due to uneven heating	Air flows from high to low pressure.
48	**(b)** melting point	Melting point.
49	**(a)** sonority	Sonorous.
50	**(a)** suspension	Large particles.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** reflects light into our eyes	Reflection.
2	**(b)** full moon	Purnima.
3	**(b)** producers	Green plants.
4	**(c)** troposphere	Where we live.
5	**(a)** curiosity and willingness to question	Science asks for evidence.
6	**(c)** a simple microscope with a single lens	He ground his own lenses.
7	**(d)** anaemia	Low haemoglobin.
8	**(b)** a thin nichrome wire	High resistance means more heating.
9	**(b)** attract each other	Opposite attract.
10	**(c)** warm oceans	Needs warm water.
11	**(c)** cooling	Fastest particles leave, so cooling.
12	**(a)** mixture of gases	Mostly nitrogen and oxygen.
13	**(a)** 100 g	Solute + solvent.
14	**(c)** concave mirror	Magnified upright image.
15	**(c)** 365 days (366 in a leap year)	Solar.
16	**(b)** oxygen	Oxygen.
17	**(a)** groundwater	Stored underground.
18	**(b)** hypothesis	A hypothesis is a testable guess.
19	**(d)** cell membrane	It is selectively permeable.
20	**(b)** diabetes	It does not spread from person to person.
21	**(b)** copper	Metals conduct electricity.
22	**(b)** friction	Friction acts only on contact.
23	**(b)** its edge has a smaller area, so pressure is greater	Same force over smaller area gives larger pressure.
24	**(a)** diffusion	Particles of ink mix with water particles.
25	**(a)** Na	From the Latin natrium.
26	**(a)** dissolves a large number of substances	It dissolves many but not all.
27	**(d)** normal	Always from the normal.
28	**(d)** 366 days	An extra day is added to February.
29	**(a)** producers	Autotrophs.
30	**(c)** inner core	Solid, very hot.
31	**(d)** organise data and see patterns	Tables show trends clearly.
32	**(c)** multicellular	Many cells make the body.
33	**(b)** disease-causing microorganisms	Pathogens cause diseases.
34	**(c)** magnetic effect of current	An electromagnet pulls the hammer.
35	**(a)** change sliding into rolling friction	Reduce friction.
36	**(c)** rises up	It expands and becomes lighter.
37	**(c)** 0 degrees Celsius	Ice melts at 0.
38	**(d)** sulphur	Non-metals are usually brittle solids.
39	**(d)** suspension	Soil settles.
40	**(a)** 3	n = 360/90 - 1.
41	**(c)** new moon	Amavasya.

42	**(b)** many interlinked food chains	Network.
43	**(b)** stratosphere	Lies above the troposphere.
44	**(d)** away from you and others	Hot liquid may spurt out.
45	**(a)** their functions differ	Shape fits function.
46	**(b)** vitamin D	Weak bones in children.
47	**(c)** magnetic effect of current	Oersted's observation.
48	**(a)** gold-leaf electroscope	Leaves diverge.
49	**(a)** hurricanes	Names depend on region.
50	**(c)** cools the body	Cooling by evaporation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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